

Comparison of Classical, Neural ODE, and Neural SDE Fits for Water Kefir Data

Abstract

This report compares three dynamical descriptions of water kefir fermentation data: a classical logistic ordinary differential equation, a deterministic Neural ODE, and a Neural SDE. The data consist of three trials measured at 15 time points, giving 45 scalar observations. The deterministic Neural ODE is treated as a population model: the three trials are interpreted as repeated realizations of the same experiment, so one fitted curve is compared with all trial observations. The reported residual criteria are in-sample diagnostics; out-of-sample MSE should be computed with a time-respecting train/test split.

1 Data and preprocessing

Let $y_i^{[j]}$ denote the response observed at time t_i in trial j , with $i = 0, \dots, T-1$ and $j = 1, \dots, J$. In this experiment $T = 15$ and $J = 3$. The time points run from 0 to 168 in the original time units. The observed index set is

$$\Omega = \{(i, j) : y_i^{[j]} \text{ is observed}\}, \quad n = |\Omega| = 45.$$

The neural models were trained on normalized time and normalized response:

$$\tau_i = \frac{t_i - t_{\min}}{t_{\max} - t_{\min}}, \quad z_i^{[j]} = \frac{y_i^{[j]} - \bar{y}}{s_y},$$

where

$$\bar{y} = \frac{1}{n} \sum_{(i,j) \in \Omega} y_i^{[j]}, \quad s_y = \left[\frac{1}{n} \sum_{(i,j) \in \Omega} (y_i^{[j]} - \bar{y})^2 \right]^{1/2}.$$

Thus $\tau_i \in [0, 1]$. All model diagnostics were computed on the original response scale; for the neural models this required transforming fitted normalized values back to the scale of $y_i^{[j]}$.

2 Models

2.1 Classical logistic ODE

The classical baseline is the Verhulst logistic growth model [9], fitted here as a shared-parameter nonlinear regression model. Unlike the neural models, it is fitted directly to the original response values $y_i^{[j]}$. The only normalization used by this baseline is the normalized time variable $\tau_i \in [0, 1]$.

Each trial has its own observed initial condition, fixed at the first measured value, but all trials share the same growth rate and carrying capacity:

$$\frac{dy^{[j]}}{d\tau} = ry^{[j]} \left(1 - \frac{y^{[j]}}{K} \right), \quad y^{[j]}(0) = y_0^{[j]} = y^{[j]}(t_0).$$

The exact solution used in the fit is

$$y^{[j]}(\tau) = \frac{K}{1 + \left(\frac{K - y_0^{[j]}}{y_0^{[j]}} \right) \exp(-r\tau)}.$$

The unknown parameters of the classical fit are only r and K ; the initial values $y_0^{[j]}$ are taken from the data and are not optimized. The fit also requires positive observations, because the closed-form logistic curve and the ratio $(K - y_0^{[j]})/y_0^{[j]}$ are only meaningful for positive initial values. Missing observations, if present, are excluded through the observed index set Ω .

The fitted parameters must satisfy $r > 0$ and $K > \max_j y_0^{[j]}$. Instead of optimizing constrained parameters directly, the implementation optimizes two unconstrained real variables $a, b \in \mathbb{R}$ and maps them to valid physical parameters with a softplus transformation:

$$r = \text{softplus}(a) + 10^{-6}, \quad K = K_{\min} + \text{softplus}(b) + 10^{-6},$$

where $K_{\min} = \max_j y_0^{[j]} + 10^{-3}$ and

$$\text{softplus}(x) = \log(1 + \exp(x)).$$

The softplus function is a smooth differentiable approximation to the positive part function. It maps every real number to a positive value, behaves approximately like $\exp(x)$ for large negative x , and behaves approximately like x for large positive x . This is useful for gradient-based fitting: Adam can update the unconstrained variables a and b freely, while the reported parameters r and K remain positive and the carrying capacity stays above every initial condition. The small constants 10^{-6} and 10^{-3} are numerical safeguards that avoid zero growth and avoid a carrying capacity equal to an initial value.

The raw parameters are initialized by applying the inverse softplus map. In the implementation, this gives an initial growth rate near 3.0 on the normalized time scale, and an initial carrying capacity slightly above the largest observed response. At each training epoch the code applies the softplus transformation and evaluates the exact logistic solution for all trials and all observation times.

The fitting criterion is the masked original-scale mean squared error

$$\mathcal{L}_{\text{logistic}}(r, K) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(y_i^{[j]} - \hat{y}^{[j]}(\tau_i) \right)^2.$$

This can also be interpreted as a maximum-likelihood fit after adding a Gaussian observation-error model on the original response scale:

$$y_i^{[j]} = \hat{y}^{[j]}(\tau_i; r, K) + \varepsilon_i^{[j]}, \quad \varepsilon_i^{[j]} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \sigma^2).$$

For fixed σ^2 , maximizing this Gaussian likelihood over r and K is equivalent to minimizing the sum of squared residuals. If σ^2 is also unknown, profiling it out gives

$$\hat{\sigma}^2(r, K) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(y_i^{[j]} - \hat{y}^{[j]}(\tau_i; r, K) \right)^2,$$

and the maximizing values of r and K are again the nonlinear least-squares solution. Therefore the implemented classical fit is a profiled Gaussian maximum-likelihood estimate under an independent additive measurement-error assumption. It is not a transition likelihood for a stochastic logistic process.

The code backpropagates through the closed-form expression above and applies an Adam update [5] to a and b . No numerical ODE solver is used for this classical baseline. This model was trained for 3000 Adam epochs with learning rate 5×10^{-2} . The fitted values were

$$\hat{r} = 7.705769, \quad \hat{K} = 47.860203.$$

Since $\tau = t/168$ for these data, the growth rate in the original time scale is approximately $\hat{r}/168 = 0.045868$ per original time unit.

2.2 Neural ODE

The deterministic neural model is the non-autonomous Neural ODE [2]

$$\frac{dz}{d\tau} = f_\theta(\tau, z), \quad z(0) = z_0.$$

The fitted solution is a single population curve, not one curve per trial. The three trials are treated as replicate observations of the same underlying trajectory. On the original response scale the initial condition is

$$y_0 = y(t_0) = \frac{1}{J} \sum_{j=1}^J y_0^{[j]},$$

the mean of the first observed values across trials. The corresponding normalized initial condition is

$$z_0 = \frac{y_0 - \bar{y}}{s_y}.$$

After solving the normalized ODE, the fitted value on the original scale is

$$\hat{y}_i = \hat{y}(t_i) = \bar{y} + s_y \hat{z}(\tau_i; \theta).$$

The same value $\hat{y}_i$ is compared with every trial observation $y_i^{[j]}$ at time t_i .

The right-hand side f_θ is a fully connected neural network with input $(\tau, z) \in \mathbb{R}^2$, two hidden layers, 32 hidden units per layer, and hyperbolic tangent activation:

$$h_1 = \tanh(W_1 x + b_1), \quad h_\ell = \tanh(W_\ell h_{\ell-1} + b_\ell), \quad f_\theta(x) = W_o h_2 + b_o,$$

where $x = (\tau, z)^T$. The network has 1185 trainable weights and biases. This count comes from the three affine layers:

$$\begin{aligned} \text{input to hidden 1:} & \quad 2 \cdot 32 + 32 = 96, \\ \text{hidden 1 to hidden 2:} & \quad 32 \cdot 32 + 32 = 1056, \\ \text{hidden 2 to output:} & \quad 32 \cdot 1 + 1 = 33. \end{aligned}$$

Thus $96 + 1056 + 33 = 1185$. The architecture was chosen as a compact smooth function approximator for a scalar time-dependent vector field. The input has only two coordinates, time and state, and the data set has only 45 scalar observations, so a very large network would make the comparison dominated by overparameterization. Two hidden layers with 32 units give substantially more flexibility than the two-parameter logistic curve while remaining small enough to train stably with weight decay. The tanh nonlinearity gives a smooth right-hand side, which is useful for ODE/SDE integration, and sigmoidal neural networks are classical universal approximators [3]. The same architecture is used for the Neural ODE and for each Neural SDE coefficient so that differences between the deterministic and stochastic neural models are not caused by different network sizes.

The ODE was solved at the observed normalized times using RK4 with step size 0.01. The empirical loss was

$$\mathcal{L}_{\text{ODE}}(\theta) = \frac{1}{n} \sum_{(i,j) \in \Omega} \left(z_i^{[j]} - \hat{z}_i(\theta) \right)^2.$$

Equivalently, in original response units this loss fits one curve $\hat{y}_i$ to all three replicate measurements at the same time. The minimizer is therefore encouraged to pass through the center of the replicate cloud, while the ODE structure enforces a smooth dynamical trajectory. Training used Adam for 1000 epochs with learning rate 10^{-2} and weight decay 10^{-4} [5].

2.3 Neural SDE

The stochastic neural model is an Ito diffusion with neural drift and diffusion coefficients, following the Neural SDE modeling idea [8, 4]:

$$dZ_\tau = \mu_\theta(\tau, Z_\tau) d\tau + \sigma_\phi(\tau, Z_\tau) dW_\tau.$$

Both the drift μ_θ and the raw diffusion network use the same architecture as the Neural ODE: two hidden layers, 32 units per hidden layer, and tanh activation. The diffusion is constrained to be positive by

$$\sigma_\phi(\tau, z) = \text{softplus}(g_\phi(\tau, z)) + \sigma_{\min}, \quad \sigma_{\min} = 10^{-3}.$$

The total trainable parameter count is 2370, one copy of the 1185-parameter network for the drift and one for the diffusion.

The SDE was trained with an Euler-Maruyama Gaussian transition likelihood [6]. For successive observations separated by $\Delta\tau_i = \tau_{i+1} - \tau_i$, the approximation is

$$Z_{i+1}^{[j]} \mid Z_i^{[j]} \sim \mathcal{N} \left(Z_i^{[j]} + \mu_\theta(\tau_i, Z_i^{[j]}) \Delta\tau_i, \sigma_\phi(\tau_i, Z_i^{[j]})^2 \Delta\tau_i \right).$$

With the variance floor used in the implementation, define

$$m_i^{[j]} = z_i^{[j]} + \mu_\theta(\tau_i, z_i^{[j]}) \Delta\tau_i, \quad v_i^{[j]} = \sigma_\phi(\tau_i, z_i^{[j]})^2 \Delta\tau_i + 10^{-6}.$$

The minimized negative log-likelihood is

$$\mathcal{L}_{\text{SDE}}(\theta, \phi) = \frac{1}{|\Omega_{\text{tr}}|} \sum_{(i,j) \in \Omega_{\text{tr}}} \frac{1}{2} \left[\log(2\pi v_i^{[j]}) + \frac{(z_{i+1}^{[j]} - m_i^{[j]})^2}{v_i^{[j]}} \right],$$

where Ω_{tr} is the set of adjacent observed transitions. Training used Adam for 1000 epochs with learning rate 5×10^{-3} and weight decay 10^{-4} [5]. After training, 1000 Monte Carlo paths were simulated to estimate predictive means and central 90 percent intervals.

3 Comparison criteria

All model comparison metrics were computed on the original response scale using the fitted mean trajectory for each model. For models that produce one curve per trial, write this fitted value as $\hat{y}_i^{[j]}$. For the single-curve Neural ODE, $\hat{y}_i^{[j]} = \hat{y}_i$ for every trial j . Let

$$\text{SSE} = \sum_{(i,j) \in \Omega} (y_i^{[j]} - \hat{y}_i^{[j]})^2, \quad \text{SST} = \sum_{(i,j) \in \Omega} (y_i^{[j]} - \bar{y})^2.$$

The root mean squared error is

$$\text{RMSE} = \sqrt{\frac{\text{SSE}}{n}},$$

and the coefficient of determination is

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}.$$

The numerical table below reports in-sample residual diagnostics: the models are trained on all observed entries in Ω , and the same entries are used to compute RMSE and R^2 . This is appropriate for checking interpolation quality, but it is not an out-of-sample generalization error. To estimate a test MSE for this time-series problem, the split should be made by time rather than by randomly sampling individual observations. Let

$$I_{\text{train}} \subset \{0, \dots, T-1\}, \quad I_{\text{test}} \subset \{0, \dots, T-1\}, \quad I_{\text{train}} \cap I_{\text{test}} = \emptyset,$$

and define

$$\Omega_{\text{train}} = \{(i, j) \in \Omega : i \in I_{\text{train}}\}, \quad \Omega_{\text{test}} = \{(i, j) \in \Omega : i \in I_{\text{test}}\}.$$

The Neural ODE is then trained by replacing Ω with Ω_{train} in $\mathcal{L}_{\text{ODE}}(\theta)$, while the test MSE is computed as

$$\text{MSE}_{\text{test}} = \frac{1}{|\Omega_{\text{test}}|} \sum_{(i,j) \in \Omega_{\text{test}}} \left(y_i^{[j]} - \hat{y}_i \right)^2.$$

For these data, a natural forecasting split is to train on early times and test on the last few time points, for example training through $t = 132$ and testing at $t = 144, 156, 168$. With only 15 time points, an alternative is a rolling leave-last-time-out protocol, such as training through $t = 120$ and testing at $t = 132$, then training through $t = 132$ and testing at $t = 144$, and so on.

For AIC and BIC [1, 7], residuals were modeled as independent Gaussian errors with common variance σ^2 . The maximum likelihood variance estimate is

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n},$$

which gives the maximized log-likelihood

$$\log \hat{L} = -\frac{n}{2} [\log(2\pi) + 1 + \log(\hat{\sigma}^2)].$$

The criteria are

$$\text{AIC} = 2k - 2 \log \hat{L}, \quad \text{BIC} = k \log(n) - 2 \log \hat{L}.$$

Here $k = p + 1$, where p is the number of fitted dynamical parameters and the additional parameter is the residual variance. Thus $k = 3$ for the classical logistic model, $k = 1186$ for the Neural ODE,

and $k = 2371$ for the Neural SDE. Initial conditions are not counted as fitted parameters because they are taken directly from the first row of the data; for the single-curve Neural ODE this means using $y_0 = J^{-1} \sum_{j=1}^J y_0^{[j]}$.

The 90 percent interval coverage is computed only for the Neural SDE, because only this model produces a predictive distribution rather than a single fitted mean curve. Let $B = 1000$ be the number of simulated SDE paths and let $Y_i^{(b)[j]}$ denote the simulated original-scale value for path b , time i , and trial j . For interval level $\gamma = 0.90$, set $\alpha = (1 - \gamma)/2 = 0.05$ and define the pointwise empirical interval

$$L_i^{[j]} = Q_\alpha \left(\{Y_i^{(b)[j]}\}_{b=1}^B \right), \quad U_i^{[j]} = Q_{1-\alpha} \left(\{Y_i^{(b)[j]}\}_{b=1}^B \right),$$

where Q_q is the empirical q -quantile. The reported empirical coverage is

$$\hat{C}_{0.90} = \frac{1}{n} \sum_{(i,j) \in \Omega} \mathbf{1}\{L_i^{[j]} \leq y_i^{[j]} \leq U_i^{[j]}\}.$$

Thus an observed data point is counted as covered if it lies between the 5 percent and 95 percent Monte Carlo quantiles for the same time and trial. The reported value 0.9333 means that 42 of the 45 observed responses fall inside their corresponding central 90 percent intervals. The mean interval width is the average of $U_i^{[j]} - L_i^{[j]}$ over the same observed entries. This is an in-sample empirical calibration diagnostic for the fitted Neural SDE, not an out-of-sample guarantee.

4 Numerical results

Table 1: Model comparison on the original response scale. Lower RMSE, AIC, and BIC are better; larger R^2 is better.

Model	RMSE	R^2	AIC	BIC	p	k
Classical logistic ODE	1.4291	0.9826	165.84	171.26	2	3
Neural ODE	1.0281	0.9910	2502.20	4644.90	1185	1186
Neural SDE	2.0312	0.9649	4933.48	9217.08	2370	2371

Table 2: Additional stochastic-model diagnostics.

Model	90 percent interval coverage	Mean interval width
Neural SDE	0.9333	7.3887

The Neural ODE has the best in-sample interpolation accuracy, with the smallest RMSE and largest R^2 . The classical logistic ODE is less flexible, but it still explains approximately 98.3 percent of the observed variance. Because AIC and BIC penalize the number of fitted parameters, the two-parameter logistic model is strongly preferred by these information criteria. The Neural SDE has the largest RMSE among the three fitted mean curves, but its 90 percent predictive interval coverage is close to the nominal level.

The AIC and BIC values for the neural models should be interpreted as parameter-penalized residual criteria for the fitted mean curves. They are not the same as comparing the SDE transition likelihood directly to an exact ODE likelihood. Their practical role here is to quantify the tradeoff between in-sample residual accuracy and model complexity.

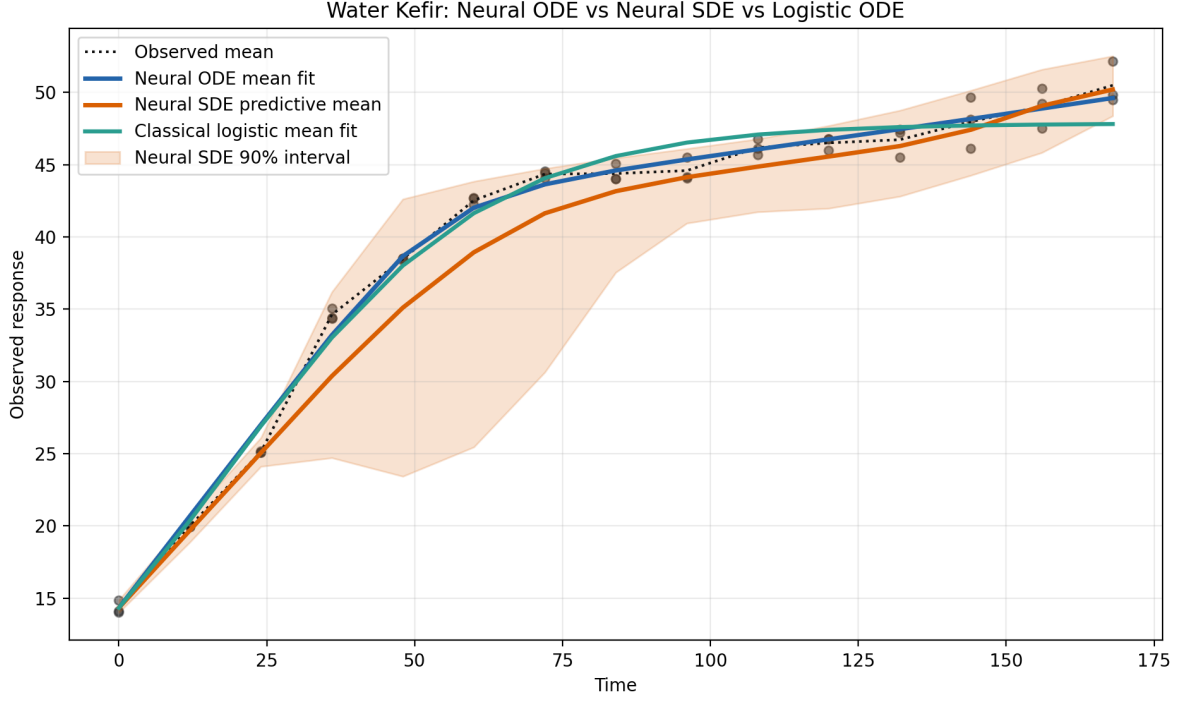


Figure 1: Observed trial data, the single Neural ODE population curve, fitted mean trajectories from the comparison models, and the Neural SDE central 90 percent predictive interval. The Neural ODE is fitted as one curve against all three trial realizations, while the logistic ODE captures the saturating growth pattern with a much smaller parameter count.

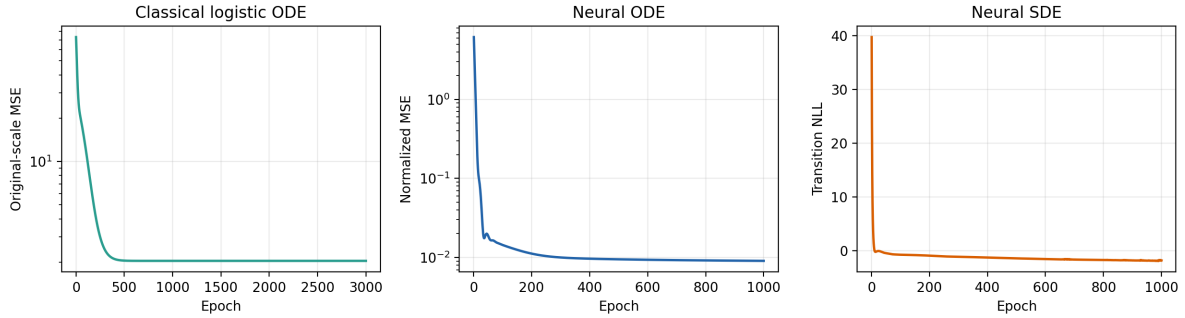


Figure 2: Training objective histories from the saved experiment outputs. The three objectives are not on the same statistical scale: the classical model uses original-scale MSE, the Neural ODE uses normalized MSE, and the Neural SDE uses normalized transition negative log-likelihood. The plots are therefore optimization diagnostics, not direct model-comparison metrics.

5 Reproducibility

The comparison was generated by the `kefir-sde-compare` command-line tool, implemented in `src/kefir_models/sde_compare.py`. The data file used in this directory is named `data/raw/waterKefirTrialsReferece.csv`. The following command reproduces the reported outputs from the project root:

```
kefir-sde-compare \
  --data-path data/raw/waterKefirTrialsReferece.csv \
  --output-dir outputs/neural_sde_comparison_outputs \
  --classical-epochs 3000 \
  --classical-learning-rate 5e-2 \
  --ode-epochs 1000 \
  --ode-learning-rate 1e-2 \
  --sde-epochs 1000 \
  --sde-learning-rate 5e-3 \
  --hidden-units 32 \
  --hidden-layers 2 \
  --weight-decay 1e-4 \
  --min-diffusion 1e-3 \
  --ode-method rk4 \
  --sde-paths 1000 \
  --interval-level 0.90 \
  --seed 7 \
  --device auto \
  --plot
```

The main output files used in this report are:

- `outputs/neural_sde_comparison_outputs/model_comparison_metrics.csv`,
- `outputs/neural_sde_comparison_outputs/classical_logistic_parameters.csv`,
- `outputs/neural_sde_comparison_outputs/water_kefir_neural_dynamics_comparison.csv`,
- `outputs/neural_sde_comparison_outputs/water_kefir_neural_dynamics_comparison.png`.

References

- [1] Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. doi: 10.1109/TAC.1974.1100705. URL <https://doi.org/10.1109/TAC.1974.1100705>.
- [2] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems 31*, pages 6571–6583, 2018. doi: 10.48550/arXiv.1806.07366. URL <https://papers.nips.cc/paper/7892-neural-ordinary-differential-equations>.

- [3] George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4):303–314, 1989. doi: 10.1007/BF02551274. URL <https://doi.org/10.1007/BF02551274>.
- [4] Patrick Kidger, James Foster, Xuechen Li, and Terry J. Lyons. Neural SDEs as infinite-dimensional GANs. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5453–5463. PMLR, 2021. doi: 10.48550/arXiv.2102.03657. URL <https://proceedings.mlr.press/v139/kidger21b.html>.
- [5] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. doi: 10.48550/arXiv.1412.6980. URL <https://arxiv.org/abs/1412.6980>.
- [6] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*, volume 23 of *Applications of Mathematics*. Springer, Berlin, Heidelberg, 1992. doi: 10.1007/978-3-662-12616-5. URL <https://doi.org/10.1007/978-3-662-12616-5>.
- [7] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi: 10.1214/aos/1176344136. URL <https://doi.org/10.1214/aos/1176344136>.
- [8] Belinda Tzen and Maxim Raginsky. Neural stochastic differential equations: Deep latent gaussian models in the diffusion limit. *arXiv preprint arXiv:1905.09883*, 2019. doi: 10.48550/arXiv.1905.09883. URL <https://arxiv.org/abs/1905.09883>.
- [9] Pierre-Francois Verhulst. Notice sur la loi que la population suit dans son accroissement. *Correspondance Mathematique et Physique*, 10:113–121, 1838. URL <https://www.scienceopen.com/document?vid=b51b58ae-889e-49ac-bc75-759cffcdc238>. No DOI located for the original nineteenth-century article.